



MILLENNIUM COLLEGE OF MEDICINE DEPARTMENT OF PUBLIC HEALTH

OUTBREAK INVESTIGATION

BY

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Module 10 Learning outcomes

- **By the end of this module, students should be able to:**
- Describe Patterns of occurrence of disease;
 - Reason out why outbreaks are investigated;
 - Describe how outbreaks are uncovered;
 - Identify Epidemic and its' types;
 - Describe the steps in the investigation of an outbreak/epidemic;
 - Describe the challenges of outbreak management.

Terminologies

Terminologies

a) Occurrence of disease at expected level includes:

- **Endemic**= disease occurrence in a population with **predictable regularity/with only minor deviations** from its expected frequency of occurrence
 - **Hyper endemic** = *affect* **high proportion** of the popⁿ at risk
 - **Meso endemic** = **moderate proportion**
 - **Hypo endemic** = **small proportion**

b) Irregular and occasional occurrence of disease:

- **Sporadic**= rarely and without **predictable regularity** of disease occurrence

Terminologies

c) Occurrence of disease at excess of the expected level

- **Epidemic** refers to the occurrence of disease or health related condition in excess of usual frequency/threshold.
- **Outbreak** is an epidemic of shorter duration covering a limited area
- **Pandemic** is an epidemic involving several countries or continents affecting a large number of people.

Definition

- An outbreak (or epidemic) is the occurrence of more cases of a disease than expected in a given area or among a specific group of people over a particular period of time.
- It is usually used interchangeably with Epidemic.

Definition

Outbreak investigations are distinct from many other types of epidemiologic studies in several important ways:

- usually **retrospective**, often relying on **recall of affected persons to identify causal linkages**
- begin **without clear hypotheses**; require descriptive studies to generate hypotheses before conducting analytic studies
- the need for responsiveness to community needs and effective risk communication is heightened
- require **public health officials** to weigh the evidence (**e.g., close a restaurant**)

OUTBREAK Ix....Why?

Why Investigate Possible Outbreaks?

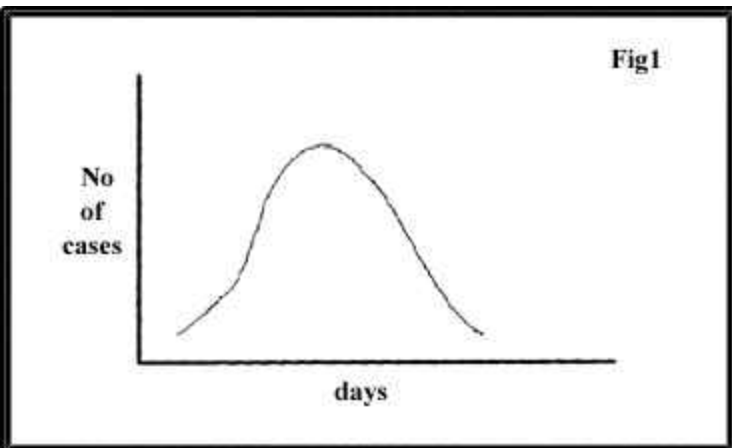
- Control and prevention measures
- Research opportunities
- Training opportunity
- Opportunity for program evaluation
- Public, political, or legal concerns

DEFINITION

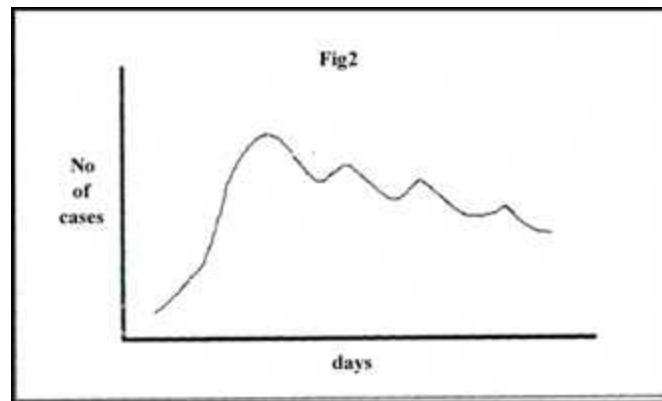
- The occurrence in a community or region of cases of an illness, specific health related behavior or other health related events clearly in excess of normal expectancy.

TYPES OF EPIDEMIC

- Common source epidemic
 - Point source
 - Common source with continuous exposure
- Propagated epidemic



Point source



propagated

STEPS OF AN OUTBREAK INVESTIGATION

1. Prepare for field work
2. Establish the existence of an outbreak
3. Verify the diagnosis
4. Define and identify cases
 - a. establish a case definition
 - b. identify and count cases
5. Perform descriptive epidemiology
6. Develop hypotheses
7. Evaluate hypotheses
8. As necessary, reconsider/refine hypotheses and execute additional studies
 - a. additional epidemiologic studies
 - b. other types of studies – laboratory, environmental
9. Implement control and prevention measures
10. Communicate findings

OUTBREAK INVESTIGATION

Composition of Field Investigation Team:

Requires team of people with skills in:

- Epidemiology
- Questionnaire design
- Interview techniques
- Biostatistics
- Data management
- Microbiology (including access to lab)

OUTBREAK INVESTIGATION

Confirm diagnosis and develop a case definition:

- Preliminary case definition is based on signs and symptoms of infection, the etiologic agent, or both
- In the definition, include period under investigation and geographic area or population in which the problem occurred.
- If the causal agent is known, it can be included in the case definition.
- As investigation proceeds, refine the case definition to increase its specificity (can classify cases as “definite”, “probable”, or “possible”).)

OUTBREAK INVESTIGATION

Confirm *diagnosis and develop a case definition*:

- Example of a typical case definition:
“All children in classroom 3 of the local school who took part in the field trip on November 20, and who fell ill with vomiting and/or diarrhea between the evening of the 20th and the evening of the 21st.”
- Usually, the case definition which start out broader than this, and become progressively more specific.

OUTBREAK INVESTIGATION

Ascertain cases:

- Conduct active surveillance to ascertain any additional cases.

In “closed” populations (e.g. day care center), may be able to identify and interview all persons potentially exposed.

In “dynamic” situations (e.g. restaurants), may need to canvass local physicians, emergency rooms, and other sources.

- Collect demographic information and means to contact each case.

OUTBREAK INVESTIGATION

Descriptive epidemiology:

Collect potential risk factor information including person, place, and time of illness

Person (age, gender, underlying illnesses)

Place (closed environment locations such as schools, sites of routine activities such as pools, and location where the person lives).

Time (onset of symptoms, activities during potential exposure period, and dates of any medical evaluations).

OUTBREAK INVESTIGATION

Construct an epidemic curve:

Plots number of cases by the onset of illness.

Provides information on:

probable time of exposure of the cases to the source(s) of infection
probable incubation period

Whether outbreak was due to common point source (e.g. contaminated food), propagated source (e.g. person-to-person contact), or both.

Whether outbreak was time limited or ongoing.

OUTBREAK PATTERNS: CLASSIFICATION OF EPIDEMICS

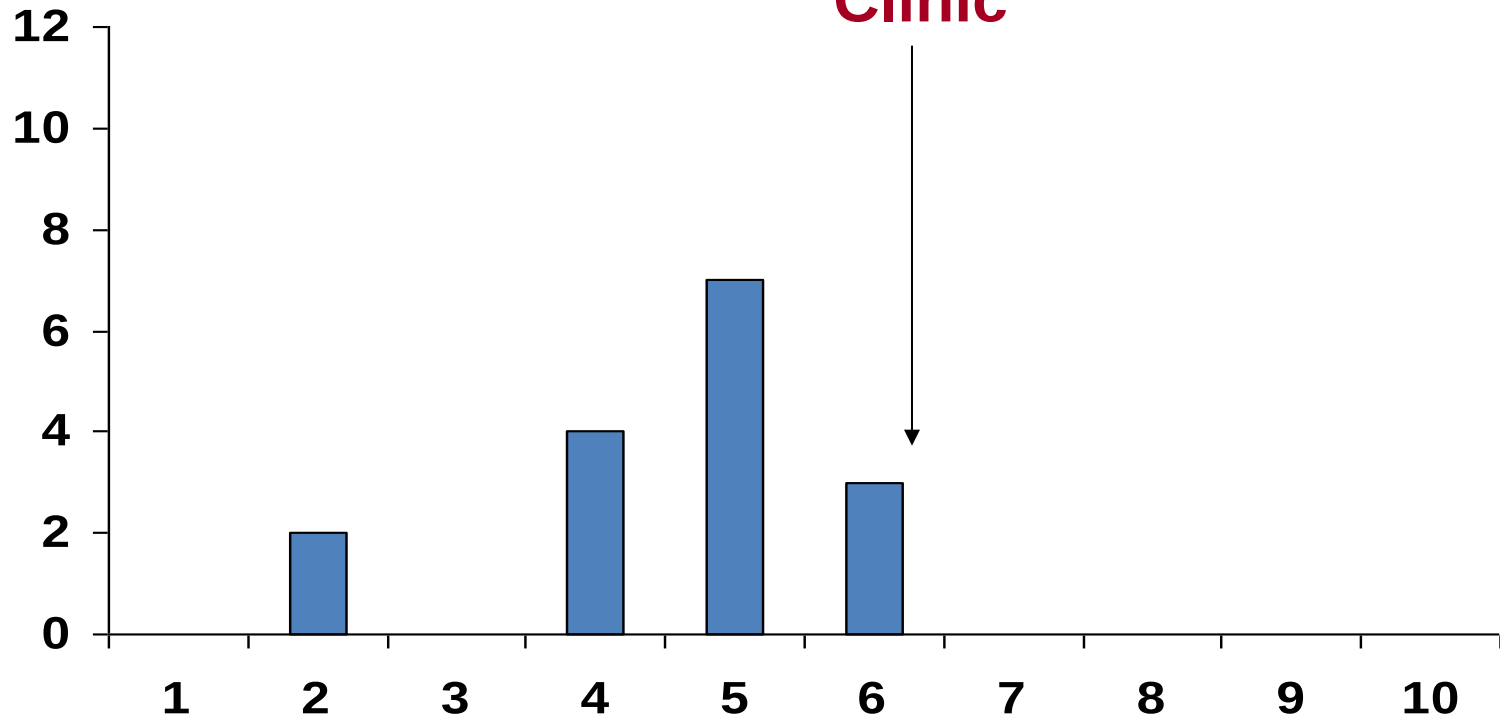
- Common Source Epidemics: all susceptible individuals have a common exposure indirect transmission most likely shape of epidemic curve
- fewest number of cases before the minimum incubation period midline of the curve defines the usual incubation period uni-modal shape typical prolonged exposure to source of outbreak widens the apex of the common source curve

OUTBREAK PATTERNS: CLASSIFICATION OF EPIDEMICS

- Point Source Epidemics
 - type of common source epidemic in which all susceptible individuals are exposed at one point in time
 - usually brief period
 - Shape of curve
 - typically sharp peak - only one incubation period
 - decline in curve is more rapid

EXAMPLE OF EPIDEMIC CURVE

of cases

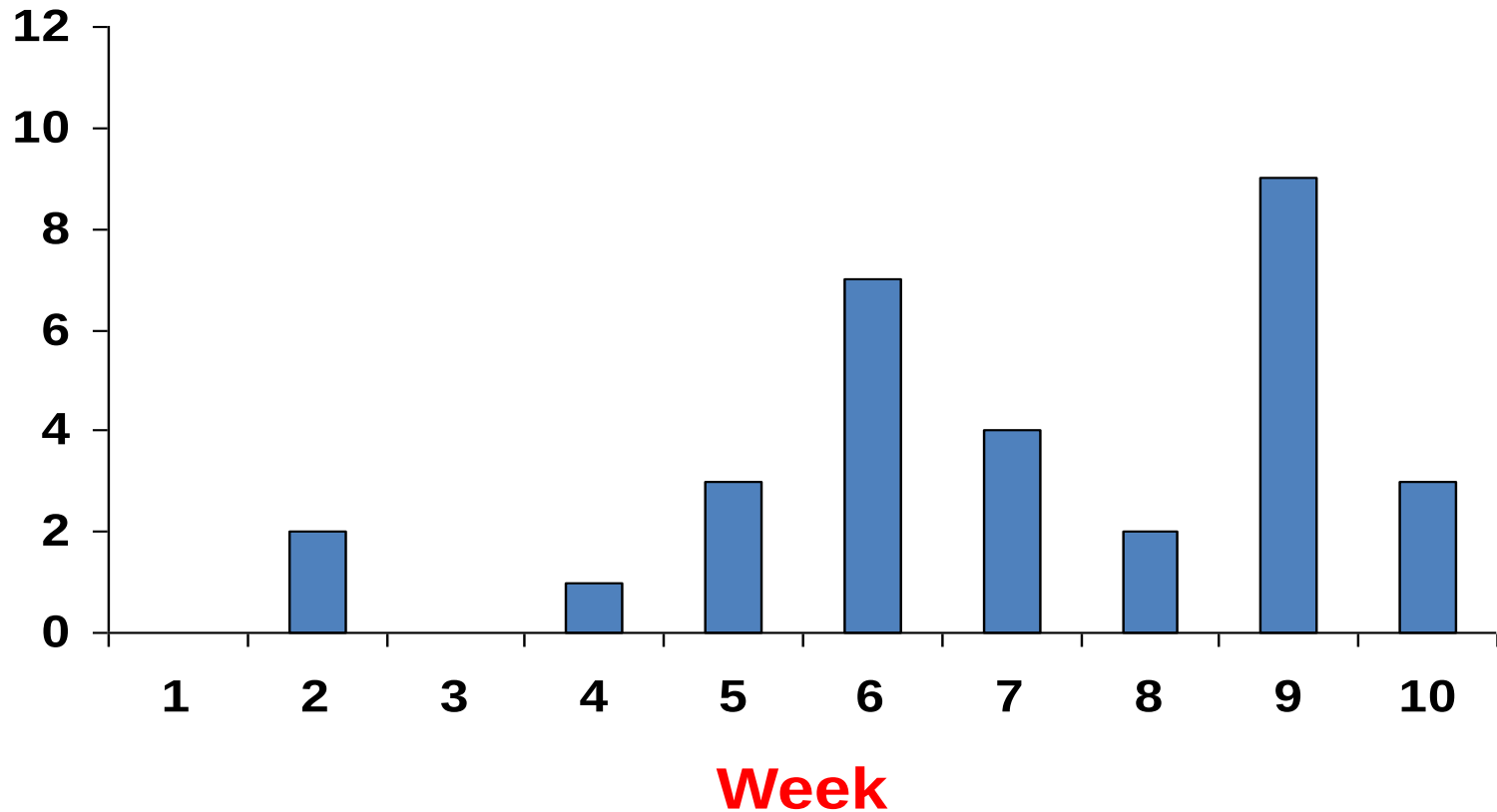


Week

Propagated source, single exposure, no secondary cases (e.g. measles)

EXAMPLE OF EPIDEMIC CURVE

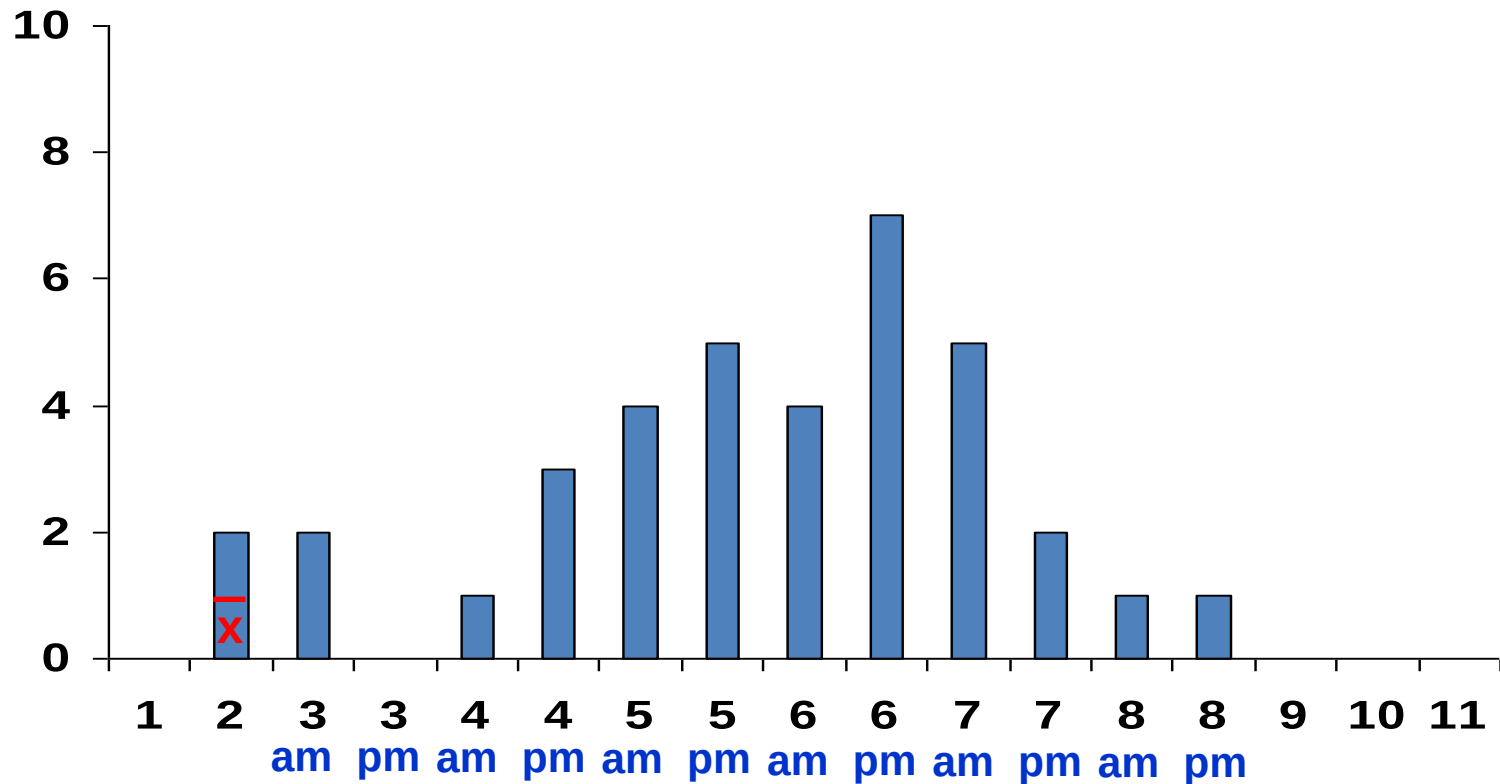
of cases



Propagated source, secondary and tertiary cases
(e.g. hepatitis A)

EXAMPLE OF EPIDEMIC CURVE

of cases



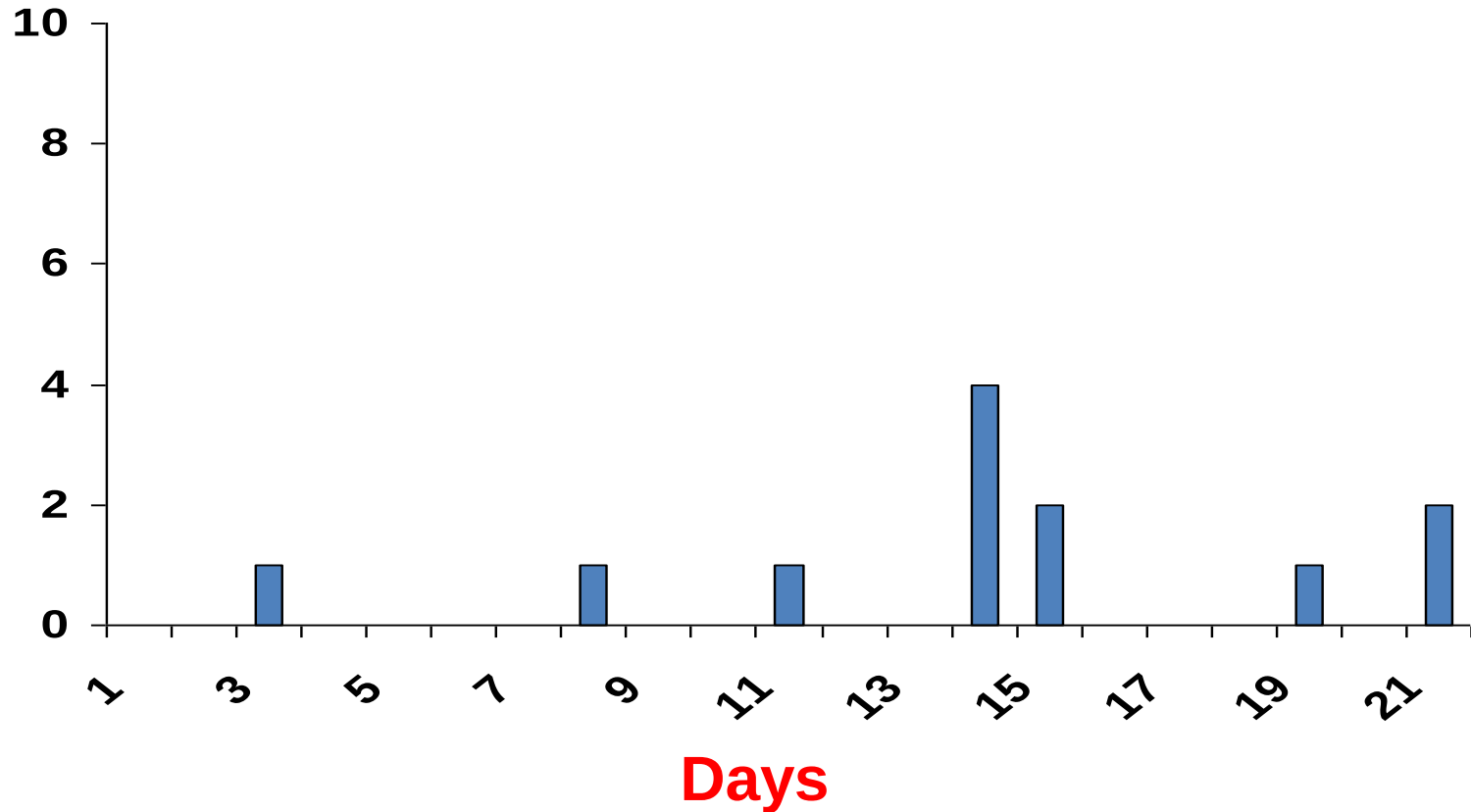
X = food handler

Days

Common source, point exposure
(e.g. salmonellosis following a company picnic)

Example of Epidemic Curve

of cases



Common source, intermittent exposure (e.g. bacteremia associated with contaminated blood product)

OUTBREAK INVESTIGATION

Formulate a hypothesis:

- Review signs/symptoms of disease exhibited by cases
GI Sx: probable ingested toxin or pathogen
Pulmonary Sx: probable inhaled toxin or pathogen
- Evaluate epidemic curve to calculate probable incubation period.
- Develop list of all potential risk factors associated with infection.
- Generate hypothesis(es) and select appropriate **study design to assess potential risk factors.**

OUTBREAK INVESTIGATION

Assess risk factors (select study design):

- In a “closed” population (e.g. hospital, school), retrospective cohort study may be used.
- In general, the case-control study is especially suited to investigating outbreaks because:
 - ✓ multiple etiologic hypotheses (exposures) can be tested concurrently
 - ✓ analysis does not require full enumeration of the whole cohort
 - ✓ makes efficient use of time and resources

OUTBREAK INVESTIGATION

Retrospective cohort study design:

- Classify persons on the basis of exposure to the suspected source of infection
- Compare “attack rate” (AR) between exposed and non-exposed persons

$$\text{AR} = \frac{\text{Number of cases of a disease}}{\text{Total population at risk for a limited period of observation}}$$

OUTBREAK INVESTIGATION

Retrospective cohort study design (example):

Of 75 persons who attended a church supper, 46 became ill within several hours ($AR = 46 / 75$): $AR = 61.3\%$

Hypothesis: Contaminated vanilla ice cream was the source of the GI infection.

Exposure status	N	Became ill	Attack rate	Relative risk
Did not eat ice cream	18	3	16.7%	1.0
Ate ice cream	54	43	79.6%	4.77

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- Note: The “secondary attack rate” (SAR) is an index of the spread of disease within a household or other circumscribed unit (all persons are exposed to primary case)

$$\text{SAR} = \frac{\text{Number of cases in group} - \text{initial case(s)}}{\text{Number of susceptible persons in group} - \text{initial cases}}$$

- SAR is conditional on being exposed to infection
- Used to assess infectivity of infectious disease agent
- May be used to evaluate efficacy of prophylactic agent

OUTBREAK INVESTIGATION

- Note: The “case fatality rate” (CFR) reflects the fatal outcome of disease, which is affected by the efficacy of available treatment

$$\text{CFR} = \frac{\text{Number of deaths due to disease X}}{\text{Number of cases of disease X}}$$

- For example, mortality from human rabies is very uncommon in the U.S. (hence low overall “mortality rate”), however, the CFR may be relatively high due to frequent failure to receive post-exposure prophylaxis.

OUTBREAK INVESTIGATION

Case control study design:

- Classify persons on the basis of infection status
- Compare odds of exposure to suspected causal agent between case and control subjects
- When outbreak occurs in a “closed” population (e.g. hospital), controls are usually selected from persons within who were potentially exposed.
- Recall bias is especially likely if the outbreak investigation is conducted after a prolonged period of time.

OUTBREAK INVESTIGATION

Case control study design (example):

Several college students presented with GI-related symptoms thought to have been associated with food served in the cafeteria.

Hypothesis: Contaminated macaroni salad was source of the GI infection.

Exposure status	Cases	Controls
Ate salad	12	4
Did not eat salad	6	14

$$OR = \frac{12 / 6}{4 / 14}$$

$$OR = 7.0$$

OUTBREAK INVESTIGATION

Conduct laboratory investigation:

- Used to isolate or otherwise identify the specific toxin or pathogen (very important for subsequent control measures)
- May be indicated when an environmental source is possible
- Molecular analyses of isolates from cases, and if available from the environment, may conclusively demonstrate presence of an outbreak and linkage to an environmental source.

OUTBREAK INVESTIGATION

Establish control measures:

- Initial control measures

Interrupt transmission from suspected reservoir

- When multiple control measures are possible, select those most effective for interrupting the epidemic, ease of implementation, expense, and safety.

Engineering changes (e.g. recall of contaminated food) are usually easier to implement than behavioral changes (e.g. alter food practices).

- Conduct follow-up studies to assess whether the incidence of illness has been reduced.

OUTBREAK INVESTIGATION

Prepare report:

- Prepared after control measures have been instituted and effectiveness has been evaluated.
- Report typically describes:
 - Extent of outbreak
 - Results of investigation
 - Control measures implemented
 - Effectiveness of control measures